COE Exploratory Data Analysis

Integrated Findings, Historical Abnormalities and Project Insights

DBA3702 Group 6 | Updated 29 October 2026

This edition places each interpretation immediately under its associated graph and incorporates the five project-level insights. A [tbc] graph marker indicates a planned visual not yet supplied. Numerical claims should be treated as descriptive EDA unless a formal validation study is cited.

1. How the datasets connect

Historical registrations and vehicle population by age identify cohorts nearing COE expiry. These feed deregistration records (vehicle exits and surrendered COEs) and COE revalidations (renewals and delayed expiry). Together with adjustments to LTA quotas, they inform the COE supply forecasting model. Historical bidding results and PQP support premium scenarios, while vehicle cost data supports dealer quotations and package comparisons. These feed a dealer decision-support dashboard for exposure, purchase timing, package evaluation and scenario analysis.

2. Important findings from the actual data

Measure	Verified finding / interpretation
238,711 cars	Cars aged 7 to under 10 in August 2026; a large approaching-expiry pool, not a guaranteed number of COEs surrendered.
32,783 renewals	Total COE revalidations in 2025 versus 26,770 in 2024. Behaviour at expiry must be explicitly accounted for.
103 quoted models	Of 217 vehicle models, 103 have nonzero COE-inclusive selling prices; only 46 have both inclusive and exclusive quotes. Zero or blank values are not free vehicles.

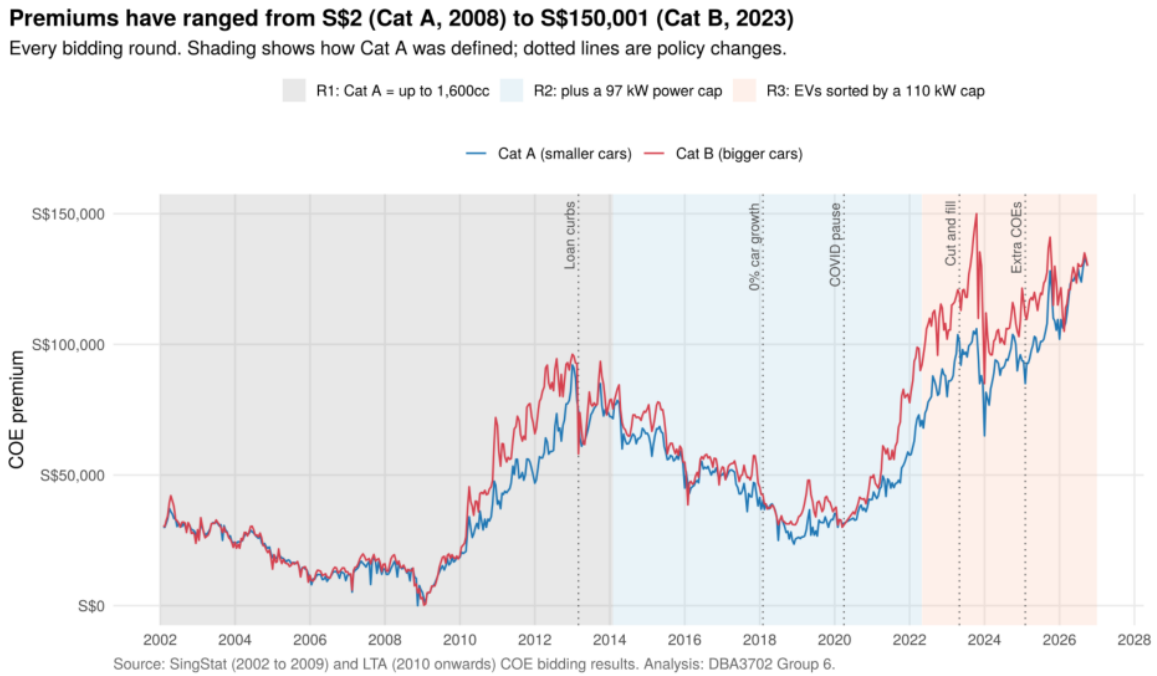
3. Vehicle cost dataset and data-quality caveats

The vehicle file has 217 models (88 Category A and 129 Category B), including 104 electric models. The basic-cost components reconcile across all rows (OMV, taxes, ARF, VES, EEAI, registration fee and COE). For the 46 dual-quoted models, the difference between inclusive and exclusive quotations is never exactly the August published premium; package comparisons therefore cannot simply add or remove the COE premium.

Issue	Caution for analysis
Historical A/B changes	Vehicle category criteria changed; avoid treating cohorts across regimes as identical.
Category E	Available in bidding data but not as a separate vehicle-population / deregistration cohort; allocation assumptions are needed.
Age data	90 missing records in the over-20-year bus category; no effect on the car-age analysis.
EV engine capacity	All 104 EVs have engine-capacity code 323, not a real 323cc engine.
Dealer costs	Median quoted selling price minus calculated basic cost is ~S$76,022 across 103 quoted models; this is not net dealer profit.

Data cross-checks: LTA and SingStat premiums agree for 1,980 rows; quota checks agree. The 2024 and 2025 revalidation totals and October 2026 round-1 published premiums were also reconciled.

Graph 1 — Category A and B COE premiums, 2002–October 2026



Project insight 4: COE premium volatility creates measurable financial exposure for dealers. In the project bidding records, changes of at least S$5,000 between the two monthly exercises occurred in approximately 18% of observed Category A months and 25% of Category B months (user-provided calculation; reproduce the month-matching procedure when presenting).

Finding. Premiums move in long cycles, ranging from Cat A S$2 in November 2008 to Cat B S$150,001 in October 2023. The shaded regimes matter: Cat A meant different things before February 2014, between February 2014 and April 2022, and after May 2022. This is not one constant car category over time.

Historical abnormalities and causes

• November 2008: Cat A cleared at S$2 in the financial crisis as demand collapsed while the quota remained large (1,852 bids for 1,851 COEs in the cited round).

• March 2013: Cat B fell from S$92,667 to S$58,090 in one round (−37%), coinciding with tighter MAS car-loan restrictions.

• 2016–2019: Prices declined as large 2005–2008 registration groups approached expiry and surrendered COEs.

• April–June 2020: No bidding during circuit-breaker suspension; this is a data gap, not a zero premium.

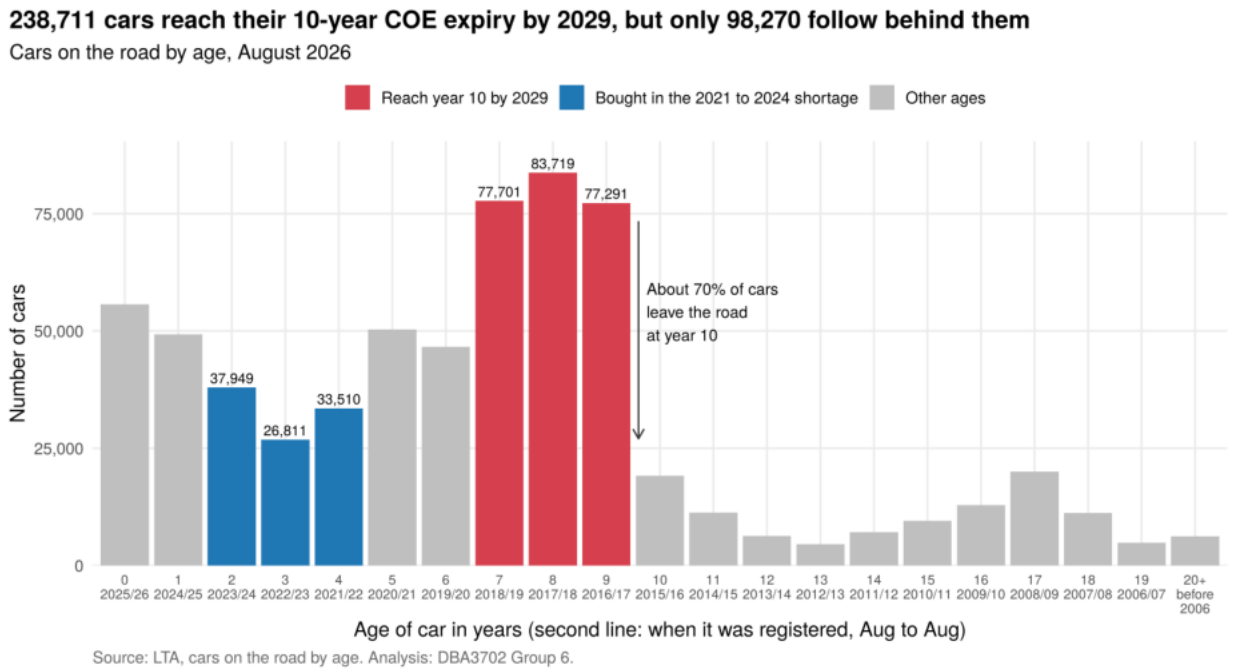
• 2021–2023: Tight supply after small earlier registration cohorts and post-COVID recovery preceded record Cat B prices.

• November 2023: Cat B fell markedly as LTA expanded available supply, illustrating policy sensitivity.

• 2026: Cat A sometimes met or exceeded Cat B. EV models tuned to Category A eligibility likely changed its competitive mix.

Dealer application. Quote-to-bid delays create material cost exposure. Track prices by category and regime, and present exposure ranges instead of assuming the next exercise repeats the previous premium.

Graph 2 — Cars on the road by age, August 2026



Project insight 1: COE supply can be anticipated from historical registration cohorts, but the precision of that forecast must be tested. The model should demonstrate incremental value over simple methods such as last-quarter quota, seasonal naive forecasts and rolling averages, using time-ordered backtesting.

Finding. There were 238,711 cars aged 7–10 nearing their year-10 decision and 98,270 cars aged 2–5 behind them. Roughly 70% of cars reaching year 10 left the road in the latest year; across ten years the estimated exit share varied from 37% to 72%. These rates do not justify treating all 238,711 vehicles as certain deregistrations.

Abnormalities and interpretation

• Large 7–10-year cohorts: 2016–2019 registration waves reflect earlier high availability as 2005–2008 cars exited.

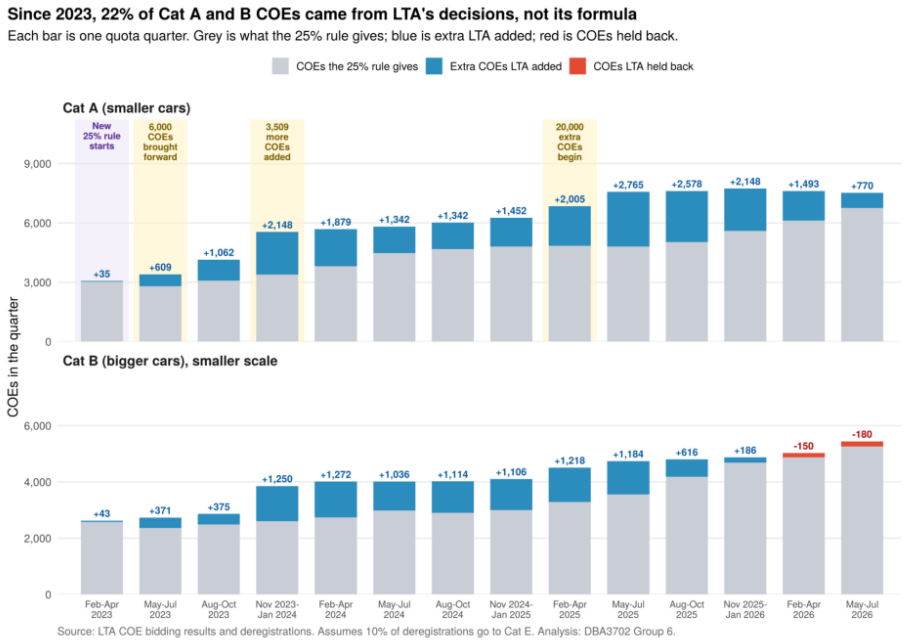
• Cliff after age 10: the age group falls sharply from 77,291 to about 19,160 cars. The 10-year COE expiry decision and PARF rebate timing encourage deregistration, though owners can renew.

• Small 2–5-year cohorts: correspond to the constrained 2021–2024 registration period, pointing to a potential supply trough around 2031–2034.

• New registrations: the youngest cohort of 55,690 cars is consistent with more recent quota expansion.

Dealer application. Use the age distribution as a leading indicator, adjust for renewal likelihood and previously brought-forward COEs, and expose prediction intervals rather than making a deterministic supply claim.

Graph 3 — Actual COE quota versus LTA baseline rule



Project insight 2: Government quota adjustments are an important part of observed supply. Separate the estimated replacement/deregistration-linked baseline from identifiable LTA additions or withholding; this is essential to interpreting model errors and building policy scenarios.

Finding. From February 2023 through the last displayed quota quarter, Cat A+B offered 142,125 COEs; estimated additional COEs under a 10% assumed Cat E share were 31,067 (22%). This is a reconstructed counterfactual based on the applicable 25% rule and Cat E allocation assumptions, not a directly published official breakdown.

Assumed Cat E share	Estimated extra A+B COEs	Share offered
5%	24,897	18%
10%	31,067	22%
15%	37,237	26%

Historical abnormalities and policy context

• February–April 2023: near-baseline quarter immediately after the 25% rule took effect. Earlier periods used different rules and should not be compared on the same basis.

• May 2023 onwards: cut-and-fill brought forward COEs from future expiry cohorts; additional volumes rose in stages.

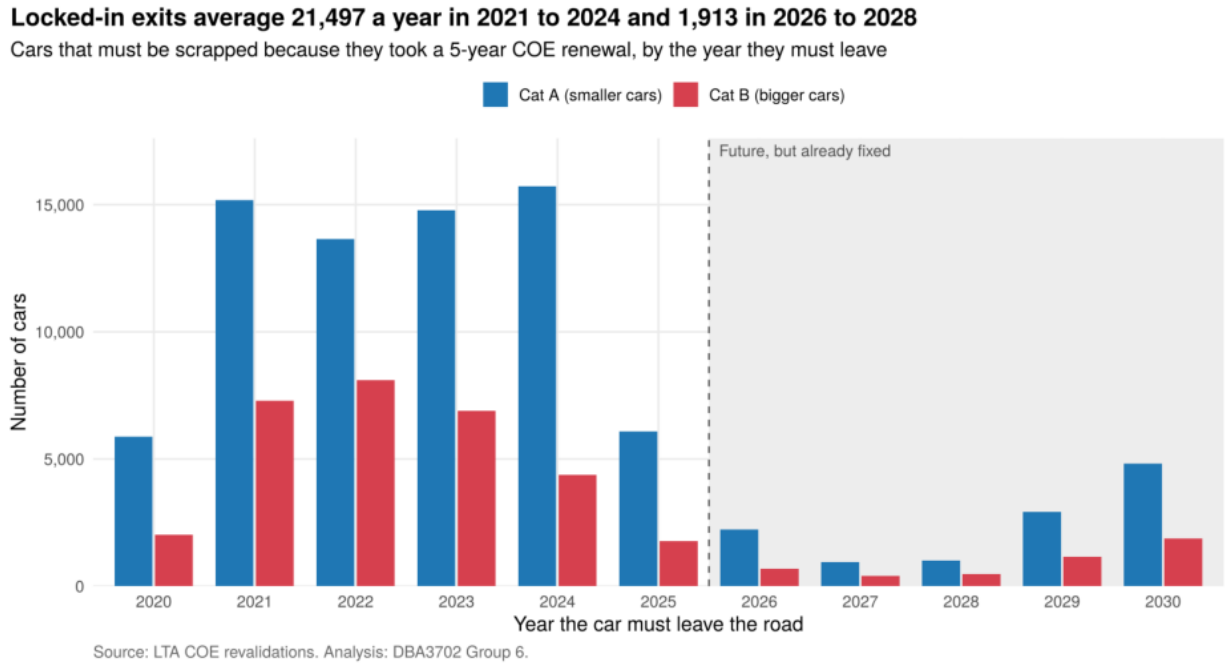
• November 2023–January 2024: Cat A +2,148 and Cat B +1,250 against the estimated rule baseline, aligned with special releases.

• From February 2025: the programme to add up to ~20,000 COEs increased observed quantities; the extra supply should be treated as a policy scenario.

• 2026: Cat B adjustments became negative versus the reconstructed baseline, consistent with supply brought forward earlier not being released twice.

Dealer application. Show a baseline and explicit government-adjustment scenarios. Do not forecast those administrative interventions solely from deregistrations.

Graph 4 — Scheduled exits after five-year COE renewals



Project insight 1: A second leading signal comes from five-year renewals, which have an identifiable final expiry date. Validate the benefit of adding this input to the historical-cohort baseline rather than merely assuming it improves the forecast.

Finding. Cars renewed for five years cannot renew for another period after that renewal ends. Potential expiry-linked exits averaged 21,497 per year in 2021–2024, but only 1,913 per year in 2026–2028. The future bars reflect scheduled end dates, not guaranteed actual exits in that year.

Abnormalities and causes

• 2021–2024: high exits from owners who took five-year renewals in 2016–2019; these renewed cars supplied a predictable expiry cohort five years later.

• 2025: sharp reduction (7,841) after a smaller preceding renewal cohort.

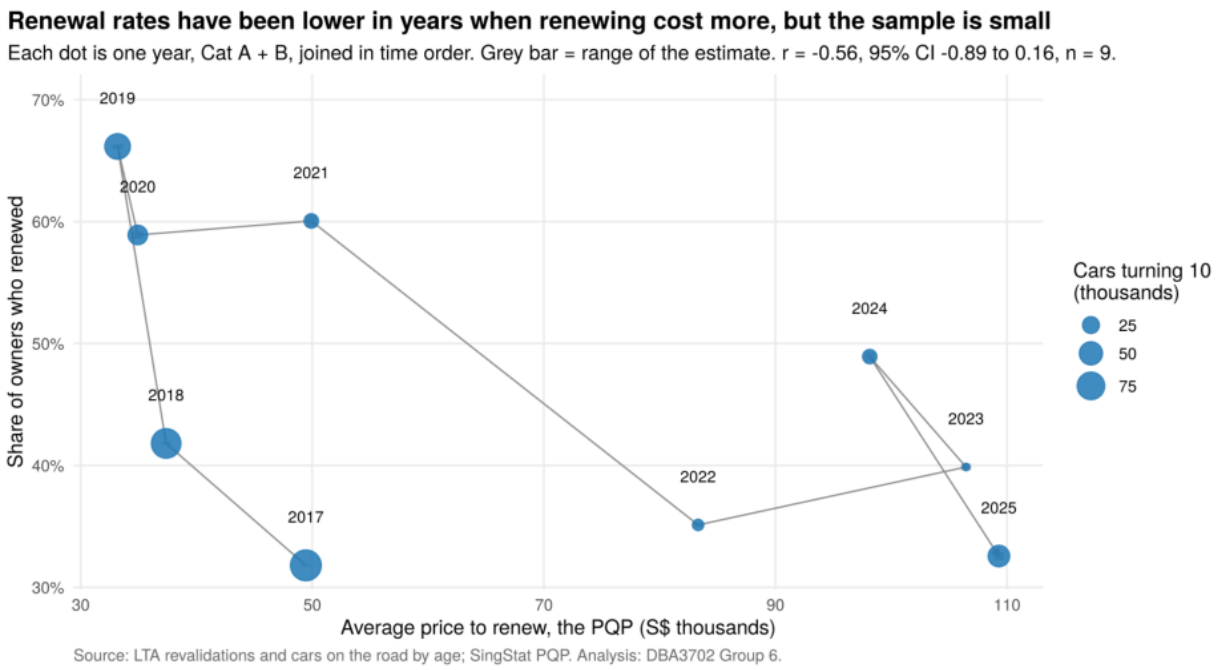
• 2026–2028: 2,912, 1,343 and 1,484 potential final expiries, respectively; the earlier tight-quota cohorts and renewal choices help explain the drop.

• 2030: rises again to around 6,680 as later renewal cohorts reach the five-year end date.

Important caveat. A car may be deregistered before its final expiry, and some associated supply may already have been front-loaded by LTA; avoid double-counting.

Dealer application. Keep fixed future expiries distinct from behavioural year-10 expiry estimates. This makes uncertainty decomposition clearer.

Graph 5 — Renewal behaviour versus prevailing PQP



Project insight 1: Renewal behaviour affects how many vehicles actually surrender COEs at year 10. Treat renewal propensity as an estimated, uncertain model input, not a fixed proportion or a proven causal response to PQP.

Corrected methodology. The renewal dataset includes cars renewing again at year 20. Dividing all renewals only by cars turning 10 therefore overstates the share renewing. The chart brackets the estimate with two denominators; these are bounds on an imperfect annual measure, not precise cohort tracking.

Finding. Across nine annual observations, renewal rate and PQP had r = −0.56 with a 95% CI of −0.89 to +0.16. Because the interval crosses zero, the negative direction is suggestive, not statistically established from this small sample. The age-series cross-check correlates PQP with the year-10 scrappage share at about +0.51.

Abnormalities and explanations

• 2017–2018: renewal rates appeared low despite relatively inexpensive PQP. Falling new-COE premiums made the three-month lagged PQP less attractive than bidding anew; 2016 MAS loan easing also improved the financing option to replace a car.

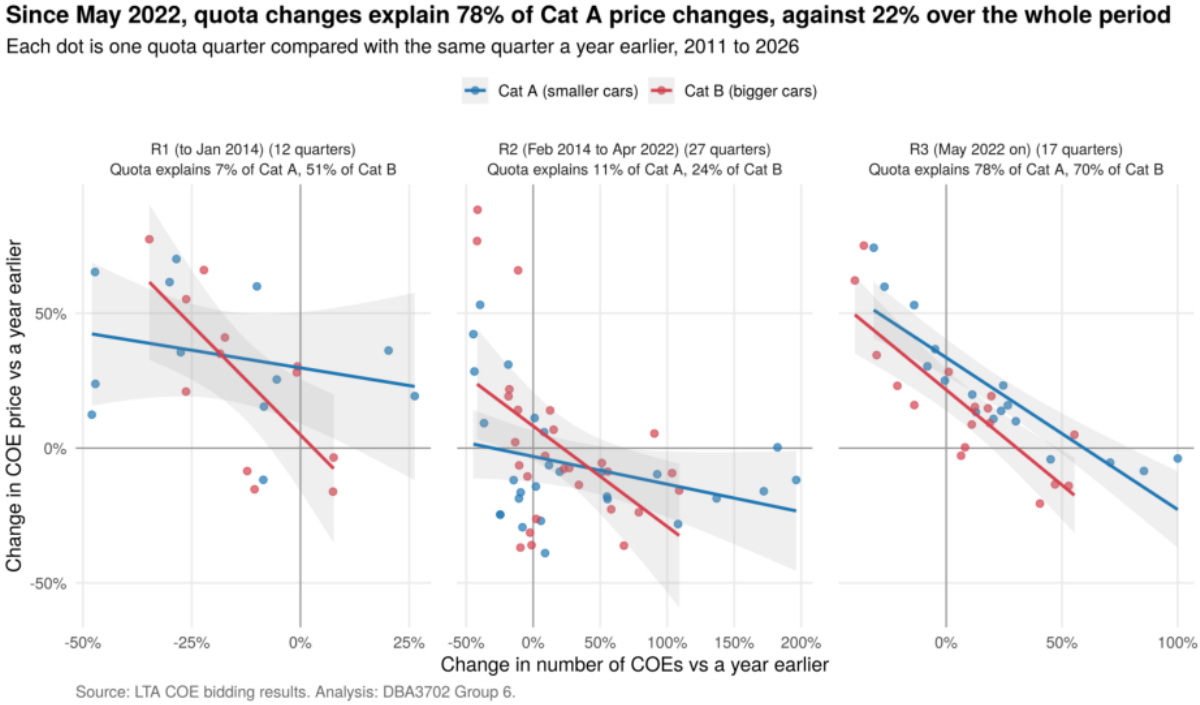
• 2019–2021: high renewal share coincided with lower PQP, while 2020 COVID uncertainty affected purchasing decisions.

• 2022–2025: renewal rates were lower at much higher PQP, although cohort sizes and other economic conditions also changed.

• PARF incentive: giving up residual PARF value at the 10-year threshold affects the renew-versus-scrap choice alongside renewal pricing.

Dealer application. Compare renewal / replacement scenarios and stress-test them. Do not simply assume high COE premiums automatically cause more renewals.

Graph 6 — Quota changes versus premium changes



Project insight 3: Higher COE supply does not guarantee lower COE premiums. Keep the supply forecast separate from premium scenarios: policy, vehicle mix and demand also affect realised prices. This distinction is a strength of the decision-support product.

Finding. Over the combined historical period, quarterly quota changes explain about 22% of the variation in year-on-year Cat A price changes and 37% for Cat B in the presented fitted relationships. Splitting by classification regime yields R^2 of 78% for Cat A and 70% for Cat B under the regime beginning May 2022.

Statistical caution. The current-regime estimate uses only 17 quarterly observations; overlapping year-on-year windows are serially dependent. An in-sample R^2 does not establish that supply is the only driver, or that the association will predict out-of-sample premiums.

Abnormalities and causes

• 2015–2016: large quota increases did not always produce proportionate price falls, partly because deregistrations can create replacement demand at the same time as they add quota.

• 2024–2026: Cat A quota increased while premiums remained high. Increased competition from EVs designed to qualify for Cat A is one plausible demand-side explanation.

• Category regime breaks: older and newer Cat A categories include different vehicle mixes, so a single pooled quota-to-price relationship can be misleading.

• Bid-to-quota ratio: not a clean demand proxy in a dynamic auction, because bids adjust in response to live clearing prices.

Dealer application. Use quota forecasts as a leading pricing signal, but show plausible premium changes and uncertainty instead of presenting a deterministic premium prediction.

Project insight 5 — Dealer decision support

Graph: [tbc]

Project insight 5: Combining forecast uncertainty with a dealer’s order book can support better pricing and timing decisions. The final business contribution is to quantify financial consequences under several supply and premium scenarios, rather than just describing historical market movements.

Suggested graph / prototype [tbc]

• Visualise the distribution of exposure for an order book, grouped by Cat A and B, comparing committed customer quotations with several future COE-price scenarios.

• Show exposure per vehicle and in total: scenario premium minus premium assumed in the quote, multiplied by unfulfilled orders, subject to the package’s contractual assumptions.

• Display timing and sensitivity analysis: next exercise versus later exercises; base, upside and downside premium assumptions; policy-adjusted quota outcomes.

• Allow dealers to compare packages using verified vehicle basic-cost components, but flag missing quotations, the fact that inclusive/exclusive selling-price differences are not simply COE, and other business costs.

Important distinction: an exposure scenario is not a forecast of an actual clearing premium. It enables a transparent stress test and should make the assumption on premiums visible.

What must be tested / confirmed

• Time-ordered out-of-sample backtests versus naive/seasonal benchmarks for the supply model (Insight 1).

• Reconciliation of the estimated baseline and official additions under several Cat E assumptions (Insight 2).

• Scenario calibration, including the historically observed monthly S$5,000 bid-to-bid changes quoted in Insight 4.

• Dealer workflow validation: whether exposure and timing information materially changes quoting or purchasing decisions (Insight 5).

Overall conclusion

Historical registration cohorts and five-year renewal obligations provide useful structural signals of future COE supply. Government decisions produce material deviations from that structure. A premium is also influenced by factors beyond supply, so the dashboard should combine validated quota forecasts, explicit premium scenarios and dealership-specific order-book exposure to support decisions with visible uncertainty.

Source basis: user-provided COE_EDA_updated_final.pdf, COE EDA Findings and Abnormalities Explained(1).pdf, and the supplementary dataset / vehicle-cost screenshots and five written insights. Historical explanations are descriptive associations, not standalone causal proof.